

# DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

## Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: IV

Subject Name: Hydraulics - II

Subject Code: BTCVC405

Max Marks: 60

Date: 25/01/2024

Duration: 3 Hrs.

### Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                   | (Level/CO)        | Marks     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                       |                   | <b>12</b> |
| A) Explain open channel flow and state difference between open channel flow and pipe flow.                                                                                                                                                                                                                                                                                        | <b>Remember</b>   | <b>6</b>  |
| B) Find an expression for the discharge over a rectangular weir using neat labelled diagram.                                                                                                                                                                                                                                                                                      | <b>Remember</b>   | <b>6</b>  |
| C) Find the time required to lower the water level from 3 m to 1.5 m in reservoir of dimension 70 m X 70 m, by (i) a rectangular notch of length 2.0 m, (ii) a right angled V notch is used. Take $C_d = 0.60$ . Take all other data same.                                                                                                                                        | <b>CO 1</b>       | <b>6</b>  |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                        |                   | <b>12</b> |
| A) Define - most economical section of channel and derive conditions for most efficient triangular channel section.                                                                                                                                                                                                                                                               | <b>CO 1</b>       | <b>6</b>  |
| B) Explain Specific energy and Specific energy curve with neat diagram.                                                                                                                                                                                                                                                                                                           | <b>Understand</b> | <b>6</b>  |
| C) Show that, for maximum discharge in a rectangular channel, the depth of flow is $2/3$ of the specific energy.                                                                                                                                                                                                                                                                  | <b>CO 1</b>       | <b>6</b>  |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                       |                   | <b>12</b> |
| A) State the gradually varied flow assumptions and derive the dynamic equation of gradually varied flow.                                                                                                                                                                                                                                                                          | <b>CO 2</b>       | <b>6</b>  |
| B) Enlist and give description about types of hydraulic jump.                                                                                                                                                                                                                                                                                                                     | <b>CO 2</b>       | <b>6</b>  |
| C) A jet of water, 10 cm diameter moves with a velocity of 25 m/s and strikes a series of flat plates fixed on the periphery of wheel. If due to impact the wheel rotates at 100 rpm, calculate (i) Force exerted by jet on plate, (ii) Work done on the plate/sec, (iii) Torque exerted on wheel if the radial distance at which jet strikes the plate and axis of wheel is 1 m. | <b>CO 3</b>       | <b>6</b>  |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                        |                   | <b>12</b> |
| A) Elaborate major component parts of Pelton wheel, Francis and Kaplan Turbine with neat sketch.                                                                                                                                                                                                                                                                                  | <b>Remember</b>   | <b>6</b>  |

**B)** A Pelton wheel turbine having a mean bucket diameter of 1.2 m is running at 1000 r.p.m. The net head is 840 m. If the side clearance angle is 150 and discharge  $Q = 0.12 \text{ m}^3/\text{s}$ . Determine power available at the nozzle and efficiency of a Pelton wheel. Take data -  $C_v = 1$ . **CO 2** **6**

**C)** Write a note on selection of turbines. **Understand** **6**

**Q. 5 Solve Any Two of the following.** **12**

**A)** How pumps are generally classified and explain in detail working of a centrifugal pump. **CO 2** **6**

**B)** Explain i) Efficiencies of pump  
ii) Reciprocating pump. **CO 2** **6**

**C)** Describe performance characteristics of centrifugal pumps. **CO 2** **6**

**\*\*\* End \*\*\***